Analytical Chemistry Cumulative Examination
October 10, 2008

Attached is a copy of a paper published about 10 years ago as a "Correspondence" in Analytical Chemistry. Please critically review it as you would if it were sent to you prior to publication for review. In other words, imagine that you are going to make a recommendation either for or against publication and, if for publication, with or without revisions. As a good reviewer, you will need to give good reasons for any recommendation you make. If you recommend publication with revisions, you will need to specify what type of revisions are needed and why you think they are needed.

CORRESPONDENCE

Liquid Sample Introduction for Matrix-Assisted Laser Desorption Ionization

Kermit K. Murray and David H. Russell*

Department of Chemistry, Texas A&M University, College Station, Texas 77843-3255

INTRODUCTION

One of the most important tools in analytical chemistry is liquid chromatography with mass spectrometric detection (LC/MS).[1-3] The development of interfaces between liquid separation and mass spectrometry is a difficult problem because the large volume flow of liquid solvent in the LC effluent is incompatible with the high vacuum requirements of a mass spectrometer. A variety of methods have been developed to overcome this problem either by splitting off a small amount of the liquid flow or by removing the solvent from the analyte before or during ionization. In the direct liquid introduction interface,[4] a liquid flow that is limited by the pumping speed of the mass spectrometer is introduced for chemical ionization (CI). The moving belt system[5] employs a continuous flexible belt that transports the sample into the mass spectrometer while the solvent is pumped away in differentially pumped stages. Continuous flow liquid probes have been developed for ionization by fast atom bombardment (FAB),[6] laser desorption multiphoton ionization (LD-MPI),[7,8] and recently for matrix assisted laser desorption ionization (MALDI).[9] A pulsed valve liquid introduction system has also been developed.[10]

A spray of particles is a good choice for liquid introduction into a vacuum because the large surface to volume ratio of aerosols is conducive to solvent evaporation.[11] Separation of the solvent and solute is achieved by beam skimmers and differential pumping stages. Aerosols can be formed by rapid heating (thermospray),[12] in a corona discharge (electrospray),[13,14] or by pneumatic or ultrasonic/pneumatic nebulization with CI or electron impact (EI) ionization.[15] A pneumatic electrospray technique, dubbed ion spray,[16] has also been developed.

In this paper initial results are presented for a new liquid introduction method for a mass spectrometer based on the application of MALDI to aerosols. Details of the method will be presented in future publications. Aerosols are formed using a continuous pneumatic aerosol generator from an acidified methanol solution containing the matrix and analyte. The solvent is removed by passing the aerosol through a heated tube to form a collimated beam of dried particles. Ions are formed by irradiating the aerosol with pulsed UV radiation in the source region of a linear TOF MS. The advantages of the aerosol MALDI TOF method are (1) liquid flow capacity of 0.5 mL/min, (2) rapid acquisition of the entire mass spectrum, and (3) ionization of large peptides and proteins. In this paper, the aerosol MALDI mass spectra of gramicidin S and lysozyme (chicken egg white) are presented. Flow injection mass spectra of gramicidin S are presented to demonstrate the feasibility of coupling aerosol MALDI with HPLC.

EXPERIMENTAL SECTION

The apparatus used in these experiments will be described in detail in a future publication[17] and will be briefly outlined here. A schematic diagram of the aerosol time of flight mass spectrometer is shown in Figure 1. A syringe pump delivers a methanol solution acidified with trifluoroacetic acid and containing matrix and analyte to the pneumatic nebulizer at a flow rate of 8 μL/s (500 μL/min). Nitrogen gas sprays the liquid into a vacuum chamber evacuated by a 330 L/s roots blower and rotary piston backing pump. Optical microscopy of collected particles indicates that aerosols prepared in this manner are approximately 50 μm in diameter *before solvent evaporation* (desolvated aerosols are smaller). The operating pressure in the aerosol formation region is approximately 1 Torr. The aerosol beam is skimmed 2 cm downstream when it enters a 25 cm long, 4 mm inner diameter drying tube that is resistively heated to 300 °C. The dried aerosols enter the ion formation region and cross the pulsed laser beam at a 90° angle. Ions formed by the UV laser pulse are extracted by 5 kV applied to a repeller plate with the extraction grid held at ground. The laser used in this study was a 355-nm frequency tripled Nd:YAG focused to a spot size of 0.5 mm in the ion formation volume. A typical 1-mJ pulse energy and 6-ns pulse duration give an energy flux of 0.5 J/cm^2 and an irradiance of 80 MW/cm^2. Mass separation is achieved in a differentially pumped 1.1-m linear flight tube with a microchannel plate detector. The ion formation chamber is evacuated by a liquid nitrogen trapped 2400 L/s 6-in. diffusion pump and the detector region is evacuated by an unbaffled 1200 L/s 4-in. diffusion pump, giving operating pressures of 1×10^{-4} and 1×10^{-5} Torr, respectively. Mass spectra are acquired with a digital oscilloscope and transferred to a microcomputer for analysis.

(1) Covey, T. R.; Lee, E. D.; Bruins, A. P.; Henion, J. D. *Anal. Chem.* **1986**, *58*, 1451–1461A.
(2) Arpino, P. J.; Guiochon, G. *Anal. Chem.* **1979**, *51*, 682–701A.
(3) Huang, E. C.; Wachs, T.; Conboy, J. J.; Henion, J. D. *Anal. Chem.* **1990**, *62*, 713–725A.
(4) Baldwin, M. A.; McLafferty, F. W. *Org. Mass Spectrom.* **1973**, *7*, 1111–1112.
(5) McFadden, W. H.; Schwarz, H. L.; Evans, S. *J. Chromatogr.* **1976**, *122*, 389–396.
(6) Caprioli, R. M.; Fan, T.; Cottrell, J. S. *Anal. Chem.* **1986**, *58*, 2949–2954.
(7) Lustig, D. A.; Lubman, D. M. *Rev. Sci. Instrum.* **1991**, *62*, 957–962.
(8) Köster, C.; Dey, M.; Grotemeyer, J.; Schlag, E. W. *Proc. ASMS Conf. Mass Spectrom. Allied Top., 38th* **1990**, 1242–1243.
(9) Li, L.; Wang, A. P. L.; Coulson, L. D. *Anal. Chem.* **1993**, *65*, 493–495.
(10) Wang, A. P. L.; Li, L. *Anal. Chem.* **1992**, *64*, 769–775.
(11) Browner, R. F. *Microchem. J.* **1989**, *40*, 4–29.
(12) Blakley, C. R.; Carmody, J. J.; Vestal, M. L. *Anal. Chem.* **1980**, *52*, 1636–1641.
(13) Dole, M.; Mack, L. L.; Hines, R. L.; Mobley, R. C.; Ferguson, L. D.; Alice, M. B. *J. Chem. Phys.* **1968**, *49*, 2240–2249.
(14) Fenn, J. B.; Mann, M.; Meng, C. K.; Wong, S. F.; Whitehouse, C. M. *Mass Spectrom. Rev.* **1990**, *9*, 37–70.
(15) Willoughby, R. C.; Browner, R. F. *Anal. Chem.* **1984**, *56*, 2626–2631.
(16) Bruins, A. P.; Covey, T. R.; Henion, J. D. *Anal. Chem.* **1987**, *59*, 2642–2646.
(17) Murray, K. K.; Russell, D. H. Submitted for publication in *J. Am. Soc. Mass Spectrom.*

0003-2700/93/0365-2534$04.00/0 © 1993 American Chemical Society

A-2

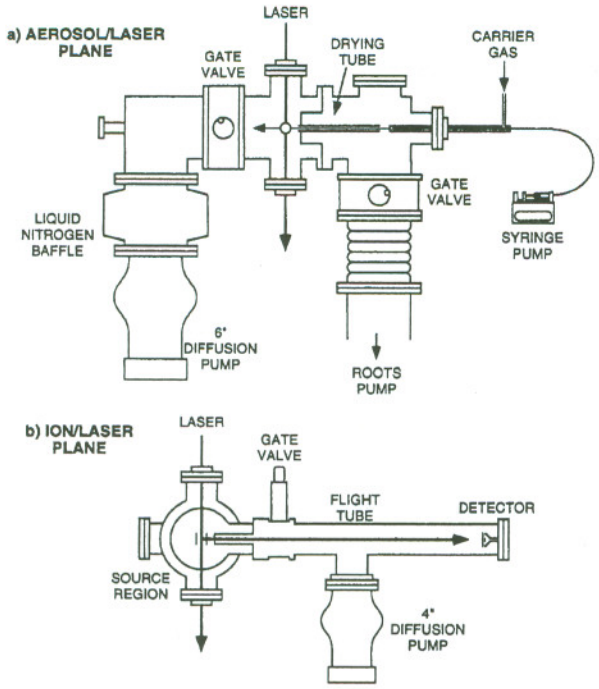

Figure 1. Aerosol MALDI apparatus. (a) View in the plane of the aerosol and laser beams. Aerosols are produced continuously in the first chamber and sent through a heated tube to the second chamber for ionization. (b) View in the plane of the laser beam and ion flight path. A 355-nm pulsed Nd:YAG laser creates ions that are accelerated for mass slection in a 1.1-m flight tube.

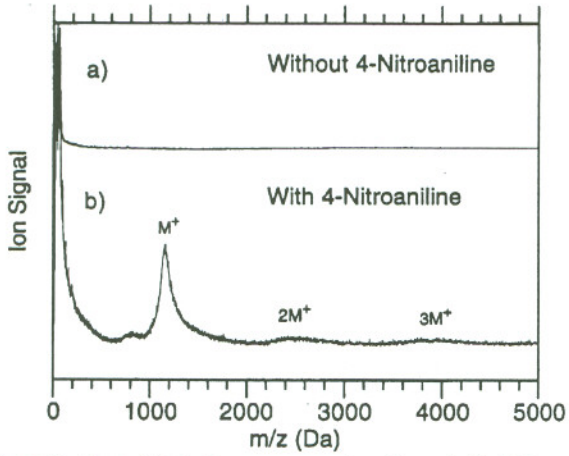

Figure 2. Aerosol ionization mass spectra of gramicidin S from an acidified methanol solution: (a) mass spectrum obtained without matrix; (b) mass spectrum obtained with 4-nitroaniline matrix.

Solutions were prepared by dissolving both the matrix and analyte in a methanol solution acidified with trifluoroacetic acid. The compounds 4-nitroaniline (99%) (Aldrich, Milwaukee, WI), gramicidin S, lysozyme from chicken egg white (Sigma, St. Louis, MO), anhydrous 99.8% spectrophotometric grade methanol (Mallinckrodt, Paris, KY), and trifluoroacetic acid (Sigma) were used without further purification.

RESULTS AND DISCUSSION

Figure 2 shows aerosol ionization mass spectra obtained from gramicidin S containing aerosols. The solution used for the mass spectrum in Figure 2a was 2.6×10^{-3} M in gramicidin S and 0.2 M in trifluoroacetic acid in a methanol solvent. The solution used for the mass spectrum in Figure 2b was 2.6×10^{-3} M in gramicidin S and 0.2 M in trifluoroacetic acid and 3.4×10^{-2} M in 4-nitroaniline. The concentrations and matrix

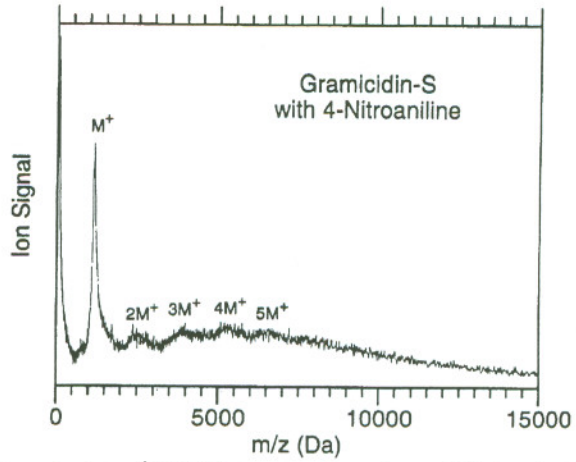

Figure 3. Aerosol MALDI mass spectrum of gramicidin S under the same conditions as Figure 1b. The mass scale is larger by a factor of 3.

to analyte ratio have not been optimized. Both spectra are averaged over 1000 laser shots at a pulse energy of 1 mJ. With no 4-nitroaniline (Figure 2a), only low mass ions (<200 Da) are detected, whereas with 4-nitroaniline added at a 13:1 molar ratio (Figure 2b), gramicidin S ions and gramicidin S cluster ions are detected. The molecular mass of gramicidin S is 1141.7 Da and the centroid of the peak labeled M^+ in Figure 2b has an m/z value of 1160 Da based on extrapolation from Na^+ and K^+ mass peaks in the spectrum (resulting primarily from impurities in the methanol solvent). It is possible that alkali metal adduct ions of the form $[M + Na]^+$ and $[M + K]^+$ are present and shift the gramicidin S M^+ peak to higher mass. Matrix-assisted laser desorption ionization of solid-state 4-nitroaniline and gramicidin S yields strong signals for $[M + H]^+$, $[M + Na]^+$, and $[M + K]^+$ ions.[18] Because gramicidin S ions are observed only when a matrix is present, it is probable that gramicidin S is ionized by H^+ transfer from 4-nitroaniline to form the $[M + H]^+$ ion as in surface MALDI.[19] The M^+ peak has a full width at half maximum of 130 Da, thus the m/z resolution of 9 is not sufficient to resolve the M^+, $[M + H]^+$, and $[M + Na]^+$ signals. It is believed that the resolution is limited by the spatial distribution of ions in the ion formation region.[17]

The mass spectrum shown in Figure 3 displays the extent of gramicidin S self-clustering in aerosol MALDI. The conditions for Figure 3 are identical to those of Figure 2b. The predominant peak in the mass spectrum occurs at the m/z value for the M^+ ion of gramicidin S. In addition, there is a distribution of clusters with a local maximum at $n = 4$. Although individual peaks cannot be resolved beyond $n = 6$, clusters up to $n = 9$ appear to contribute to the high mass tailing to baseline.

A three-dimensional plot for flow injection of gramicidin S is shown in Figure 4. The plot consists of 62 individual mass spectra, each an average of 100 laser shots saved consecutively every 10 s as they are acquired. The data were averaged by the oscilloscope and reduced to 2000 data points per mass spectrum for transfer to the microcomputer. The data were smoothed for clarity in the three-dimensional plot. A methanol solution of 4-nitroaniline at 0.014 M concentration was run continuously into the nebulizer. Three 250-μL plugs of a mixture of gramicidin S at 2.6×10^{-3} M concentration and 4-nitroaniline at 0.014 M concentration were injected into the flow at 2, 5, and 8 min. The TFA concentration in

(18) Gimon, M. E.; Russell, D. H. Unpublished results.
(19) Hillenkamp, F.; Karas, M.; Beavis, R. C.; Chait, B. T. *Anal. Chem.* 1991, *63*, 1193–1202A.

A-3

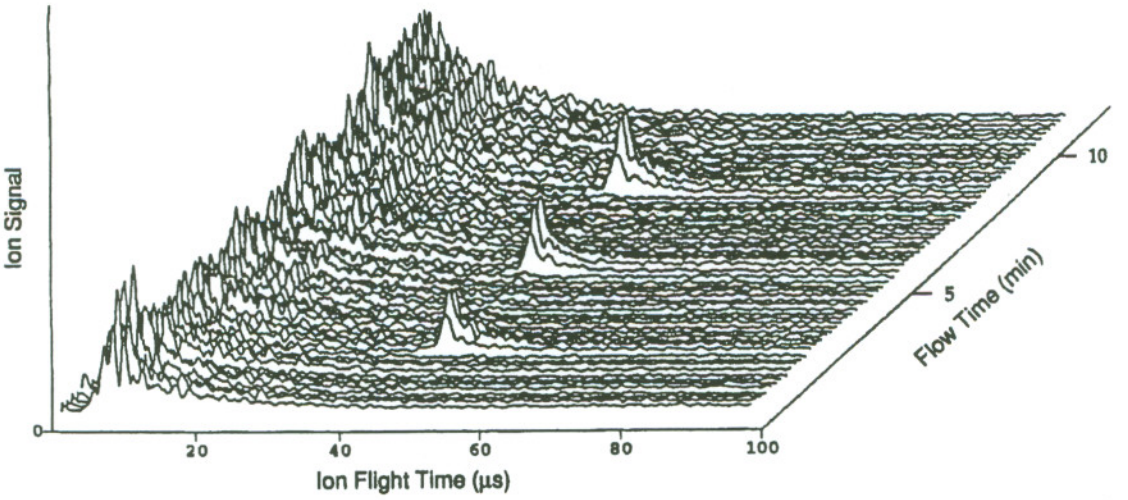

Figure 4. 3-D flow injection spectrum of gramicidin S (MW = 1141.7) in a 4-nitroaniline matrix. Three plugs of 250 μL each were injected into the mass spectrometer at 3-min intervals.

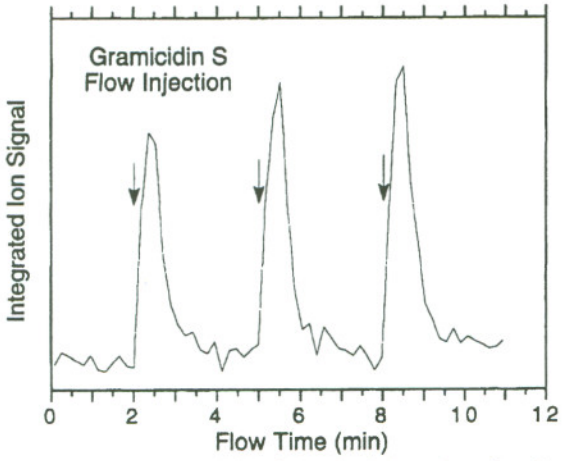

Figure 5. Integrated gramicidin S ion signal plotted as a function of time in the flow injection experiment. The arrows correspond to the times of the three gramicidin S injections.

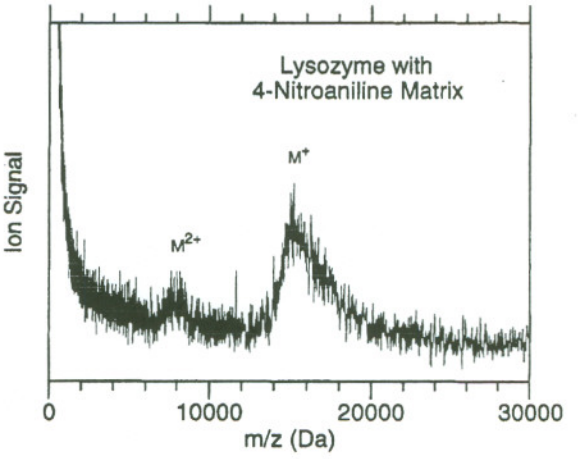

Figure 6. Aerosol MALDI mass spectrum of lysozyme (MW = 14307) obtained with 4-nitroaniline matrix at a 60:1 molar ratio.

the solutions was 0.2 M. The three resulting gramicidin S peaks are seen at approximately 45-μs flight time, rising after the inject time. The gramicidin S single ion "flowgram" extracted from Figure 4 is shown in Figure 5. The arrows correspond to the times of the three gramicidin S injections. The three flowgram peaks were fit to Gaussian profiles and the resulting widths were 29.5, 31.2, and 32.1 s in order of increasing time. The expected peak width for the 250-μL injection is 30 s, thus peak spreading does not appear to be a problem. The peak heights increase with successive injections: 18% between the first and second injections and 9% between the second and third injections. The increasing width and height of the flowgram peaks with flow time indicate the possibility of memory effects, however, contribution due to drift of other system parameters such as laser power and drying tube temperature cannot be ruled out.

The aerosol MALDI mass spectrum of lysozyme (chicken egg white, MW = 14 307) was investigated to test the generality of the aerosol MALDI method and the applicability to peptides and proteins larger and more difficult to ionize than gramicidin S. The aerosol MALDI mass spectrum of lysozyme in a 4-nitroaniline matrix is shown in Figure 5. The matrix to analyte ratio was approximately 60:1 (this is a lower bound since the lysozyme did not fully dissolve in the solution). The concentration of 4-nitroaniline in the solution was 0.018 M and trifluoroacetic acid was 0.2 M. The TOF acceleration voltage was 5 kV and the laser pulse energy was 6 mJ at 355 nm. A mass spectrum of 700 laser shots at 25 000 data points with a 50 ns/point spacing was accumulated and reduced to 5000 with averaging for the plot in Figure 6. Prominent in the spectrum are the peaks corresponding to singly and doubly charged lysozyme, labeled M^+ and M^{2+} respectively. The M^+ and M^{2+} peaks occur at 15 900 and 8200 Da based on mass calibration using the Na^+ and K^+ peaks in the spectrum. The mass resolution assuming that the M^+ peak is a single peak is 5; however the resolution for the lower mass ions is approximately 30. The discrepancy between the mass of lysozyme and the measured peak centroid is due either to clustering that shifts the peak centroid to higher mass or to errors from extrapolation from Na^+ and K^+ to high mass.

CONCLUSIONS

We have developed a new liquid introduction method for mass spectrometry that combines matrix-assisted laser desorption ionization (MALDI) with aerosol sample introduction. The technique has the following properties: (1) HPLC compatible liquid flow rates of 0.5 mL/min can be used; (2) peptides with masses of over 10 000 Da have been ionized and detected; (3) successive mass spectra have been acquired

(20) Ens, W.; Mao, Y.; Mayer, F.; Standing, K. G. *Rapid Commun. Mass Spectrom.* **1991**, *5*, 117–123.

A-4

every 10 s; (4) a limit of detection at the 100 nmol level has been achieved; (5) mass resolution of between 5 and 10 is observed for peptides. It is reasonable that molecules as large as have been observed with surface MALDI can be observed with aerosol MALDI as well. Both the detection limit and data acquisition rate can be improved by using a higher repetition laser and faster data acquisition. Pulse counting with a time to digital converter can also improve the detection limit.[20] We believe that the major factor affecting the mass resolution is the spatial extent of the ions formed in the source. In this case, mass resolution can be improved by using a tighter laser focus and by reducing the electric field in the extraction region with multiple stage acceleration. Improved mass resolution will facilitate studies of the initial energy spread of the aerosol MALDI produced ions.

Aerosol MALDI shows promise as a general LC/MS interface for HPLC of biomolecules. It appears that many of the desirable qualities of MALDI and TOF mass analysis can be used with on-line liquid sample introduction. We are currently implementing many of the improvements described above for increased sensitivity, resolution, and scanning rate. Preliminary studies with peptides separated by HPLC show that there are no major obstacles for the execution of MALDI LC/MS.[21]

We are currently exploring some of the fundamental aspects of the aerosol MALDI process. Relevant issues include the effects of droplet size, degree of desolvation, and microcrystalline structure on the type and quantity of ions produced. The ionization efficiency in the presence of various solvents and additives is especially important for LC/MS applications. We have begun investigations to determine the optimum matrix to analyte ratios for various matrix and analyte combinations and preliminary work has been done on the laser energy dependence near the threshold for ion formation.[22] The results of these investigations show promise to elucidate the mechanisms of both the aerosol and surface MALDI processes.

ACKNOWLEDGMENT

Funding for this research was provided by the Texas Advanced Technology Program and by the Division of Chemical Sciences, Office of Basic Energy Sciences, U.S. Department of Energy under Grant No. DE-FG05-85ER-13434. The Nd:YAG laser and additional equipment were purchased with funds provided by the National Science Foundation under Grant No. CHE-8821780.

RECEIVED for review April 29, 1993. Accepted June 30, 1993.

(21) Murray, K. K.; Russell, D. H. *Proc. ASMS Conf. Mass Spectrom. Allied Top., 41st*, in press.
(22) Murray, K. K.; Russell, D. H. *Proc. Second Int. Conf. Laser Ablation, 2nd*, in press.

A-5